

Making INDUSTRIAL EMBEDDED SYSTEMS More SECURE For CLOUD And IoT APPLICATIONS



While the benefits of Industry 4.0 are aplenty, the risks cannot be ignored. There are practices one can take to minimise such risks. This article takes a look at some such solutions



Xavier Bignalet is marketing manager - Secure Products Group, Microchip Technology

With recent advances in manufacturing such as automation, remote manufacturing and control concepts, we are entering the age of Industry 4.0. While the benefits are immense, this also poses certain risks. For example, if a factory is connected to an off-site data centre through a public or private cloud, network security is a concern. A connected factory exposes machinery to the risk of unwanted access, which could lead to machines getting corrupted. At the same time, it impacts production, thereby directly

compromising company revenue.

There are some common practices one can use to minimise such risks. First, irrespective of a wired, wireless or combination solution, it is essential to ensure that the chosen connectivity technology has standard protocols (such as Wi-Fi, Bluetooth and the like). Also, while the industrial segment traditionally prefers proprietary solutions, it makes sense to use standard practices since these run a lower risk of being compromised.

One of the biggest challenges in main-